THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A student finds that a colourless liquid turns blue litmus red and does not affect red litmus. It also fizzes with baking soda. The liquid is: 1

(a) neutral **(b)** a base **(c)** a salt **(d)** an acid

2. A bulb rated 6 V is connected to a battery of three 1.5 V cells in series. The bulb will: 1

(a) fuse **(b)** glow dimly **(c)** not glow at all **(d)** glow normally

3. A shiny solid X conducts electricity, is hard, and its oxide turns red litmus blue. X is probably: 1

(a) carbon **(b)** sulphur **(c)** magnesium **(d)** phosphorus

4. A student heated a blue crystalline solid. It turned white and drops of liquid appeared on the cooler part of the tube. On adding water to the white powder, it turned blue and became warm. This indicates: 1

(a) burning **(b)** an irreversible change

(c) a physical change **(d)** a reversible chemical change

5. A 14-year-old girl is shorter than her 14-year-old male cousin. The best explanation is: 1

(a) she is old

(b) she is ill

(c) growth rates vary and the cousin may have started his spurt

(d) she is not eating

6. Two rods A and B of equal size are heated at one end. A's pins fall in 20 s, B's pins fall in 60 s. This shows that: 1

(a) A is a better conductor **(b)** B is a better conductor

(c) both equal **(d)** neither conducts

7. A train 150 m long crosses a platform 250 m long in 20 s. The speed of the train is: 1

(a) 8 m/s **(b)** 7.5 m/s **(c)** 12.5 m/s **(d)** 20 m/s

8. A person has a blocked bile duct. The food whose digestion will be most affected is: 1

(a) protein-rich food **(b)** starch-rich food

(c) fat-rich food **(d)** water

9. A leaf is kept in the dark for 48 hours, then half of it is covered with black paper and kept in sunlight for 6 hours. Which statement is correct? 1

(a) the covered half turns blue-black

(b) the covered half remains brown with iodine

(c) both turn blue-black

(d) both remain brown

10. A ray of light falls on a plane mirror along the normal. The ray: 1

(a) passes through the mirror **(b)** travels along the mirror surface
(c) is absorbed **(d)** returns along the same path

11. If the Earth's axis were tilted by 45° instead of 23.5°, the seasons would be: 1

(a) milder **(b)** absent **(c)** more extreme **(d)** the same

12. In an experiment, 20 mL of a base is neutralised by 20 mL of a certain acid of the same type. If 40 mL of the same base is to be neutralised, the acid needed is: 1

(a) 40 mL **(b)** 80 mL **(c)** 10 mL **(d)** 20 mL

13. A student claims: 'Pure water conducts electricity well'. How would you test and correct this? 1

(a) heat it
(b) water is a good conductor
(c) add sugar to make it conduct
(d) test with a tester; add salt to water to make it conduct

14. Which of these metals will give a vigorous reaction with dilute HCl? 1

(a) silver **(b)** gold **(c)** copper **(d)** magnesium

15. Which of these is the most reliable test to decide whether a change is chemical? 1

(a) change in shape **(b)** change in position
(c) formation of a new substance **(d)** change in size

16. A boy's voice is 'cracking'. This is temporary because: 1

(a) voice is lost **(b)** he is ill
(c) he is cold **(d)** larynx is growing and adjusting

17. At a seaside at noon, the sea breeze blows. At 11 p.m. the breeze direction is: 1

(a) reversed **(b)** upward **(c)** zero always **(d)** same

18. Two cyclists A and B start from the same point. A goes at 15 km/h and B at 20 km/h. After 3 h, B is ahead of A by: 1

(a) 5 km **(b)** 15 km **(c)** 35 km **(d)** 60 km

19. A man's heart rate is 80 beats per minute. In 24 hours, it beats: 1

(a) 11,520 times **(b)** 80 times **(c)** 1,15,200 times **(d)** 1,152 times

20. Which of these energy changes occurs in photosynthesis? 1

(a) light energy to chemical energy **(b)** heat to light
(c) sound to light **(d)** chemical to light

21. During a solar eclipse, the Moon's shadow falls on the Earth. The region with total darkness is the: 1

(a) halo **(b)** umbra **(c)** corona **(d)** penumbra

22. A year in which the Earth makes exactly 366.25 days in an orbit would need: 1

(a) no leap years **(b)** fewer leap years
(c) same **(d)** more frequent leap years

23. A rainwater sample turns blue litmus red. The rain may be: 1

(a) acidic **(b)** basic **(c)** neutral **(d)** salty

24. A student sets up a tester with a cell, a bulb and two wires with free ends. The bulb glows when a steel spoon is placed across the ends. The spoon is: 1

(a) a fuse **(b)** a conductor **(c)** a cell **(d)** an insulator

25. Which metal is used to coat iron sheets to protect them from rusting? 1

(a) sulphur **(b)** zinc **(c)** coal **(d)** wax

26. A shop owner stores iron nails in a damp room. After some days, nails are covered with flaky reddish layer. The best advice to prevent this: 1

(a) store in water **(b)** store in dry place or oil them

(c) store in salt water **(d)** store in vinegar

27. A graph of height against age for a typical teenager shows: 1

(a) a fall **(b)** a straight line only

(c) a rapid rise during the growth spurt **(d)** no change

28. Which of the following shows a correct application of convection? 1

(a) sea breeze **(b)** spoon heating **(c)** cloth drying **(d)** sunlight on skin

29. A pendulum was timed for 10 oscillations: 18.2 s, 18.0 s, 18.4 s. The best estimate of T is: 1

(a) 0.55 s **(b)** 1.82 s **(c)** 1.8 s **(d)** 18.2 s

30. The reason that the human small intestine is long and coiled is: 1

(a) to store food

(b) to make a noise

(c) to digest water

(d) to provide a large surface area for absorption

31. A farmer notices that his legume plants in poor soil have swollen nodules in roots. This indicates: 1

(a) fungi attack **(b)** water logging

(c) nitrogen-fixing bacteria **(d)** insects

32. Why does the image in a mirror appear to be as far behind as the object is in front? 1

(a) virtual image position follows the law of reflection

(b) mirror is smaller

(c) light bends

(d) mirror is thick

33. A student claims that the Earth's shadow is formed only during lunar eclipses. The correct statement is: 1

(a) it is formed by the Moon

(b) the Earth always has a shadow but the Moon enters it only during eclipses

(c) the shadow forms only then

(d) no shadow exists

34. A liquid turns turmeric red. When liquid Y is added, the colour changes back to yellow. Liquid Y is: 1

(a) an acid **(b)** water **(c)** a base **(d)** sugar

35. A student connects five 1.5 V cells in series to a 6 V bulb. The bulb will: 1

(a) not glow **(b)** glow normally

(c) glow very bright and may fuse **(d)** be dim

36. A yellow brittle solid Y burns with a blue flame and its oxide turns blue litmus red. Y is: 1

(a) sulphur **(b)** copper **(c)** magnesium **(d)** iron

37. An iron nail kept in blue solution X gets reddish-brown coating and the solution turns green. X is likely: 1

(a) copper sulphate **(b)** sugar **(c)** sodium chloride **(d)** lime water

38. A teenager thinks late puberty means something is wrong. The correct advice: 1

(a) it is rare **(b)** it is within the normal range
(c) it is a disease **(d)** it must be treated

39. A student places a black can and a white can with equal water in the sun. After 30 min the black can's water is hotter because: 1

(a) it reflects more **(b)** it absorbs more radiation
(c) it has more water **(d)** it conducts less

40. A train moving at a steady 15 m/s runs for 20 s. The distance covered is: 1

(a) 300 m **(b)** 75 m **(c)** 0.75 m **(d)** 35 m

41. A person has no pancreas. Which will be the most affected? 1

(a) digestion of water only **(b)** absorption of oxygen
(c) digestion of all three food types **(d)** breathing

42. A variegated leaf kept in sunlight: only the green part turns blue-black. This shows: 1

(a) chlorophyll is needed **(b)** CO2 is not needed
(c) water is not needed **(d)** light is not needed

43. If the angle of incidence of a ray on a plane mirror is increased by 15°, the angle of reflection: 1

(a) increases by 15° **(b)** stays the same **(c)** doubles **(d)** decreases by 15°

44. If the Earth rotated twice as fast, the day would be about: 1

(a) 6 hours **(b)** 48 hours **(c)** 12 hours **(d)** 24 hours

45. In a neutralisation, if 15 mL of acid is needed for 25 mL of a base, then for 50 mL of the base it will be: 1

(a) 60 mL **(b)** 25 mL **(c)** 30 mL **(d)** 15 mL

46. A tester lights when connected across tap water but only dimly across distilled water. Reason: 1

(a) tap water has dissolved salts **(b)** tap water is pure
(c) distilled water has salts **(d)** distilled water is wet

47. In $\mathrm{Fe} + \mathrm{CuSO_4} \longrightarrow \mathrm{FeSO_4} + \mathrm{Cu}$, the metal that displaces copper is: 1

(a) Cu **(b)** $\mathrm{CuSO_4}$ **(c)** Fe **(d)** $\mathrm{FeSO_4}$

48. Which of the following would you use to show that a new gas is formed? 1

(a) balance **(b)** ruler **(c)** magnet **(d)** lime water test

49. If the ovaries do not function, a girl may not have: 1

(a) breathing **(b)** digestion **(c)** menstruation **(d)** growth

50. At the junction between the two breezes at around dusk, the wind: 1

(a) reverses at once strongly **(b)** doubles
(c) becomes a storm **(d)** nearly stops

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND